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KOLHAPUR INSTITUTE OF TECHNOLOGY'S
COLLEGE OF ENGINEERING (AUTONOMOUS), KOLHAPUR
(AFFILIATED TO SHIVAJI UNIVERSITY, KOLHAPUR)

S.Y. B.Tech. (Computer Science and Business Systems)

(Semester- IV)

MAKEUP EXAMINATION, AUGUST- 2024

Course Code: **UCBC0401**

Course Name: **Operating Systems**

Day and Date: **Tuesday , 21-Aug-24** **PRN:**

Time: **02:00PM To 05:00PM** **Max Marks:** **100**

Instructions:

IMP: Verify that you have received question paper with correct course, code, branch etc.

i) All questions are compulsory.
ii) Figure to the right indicate full marks.
iii) Assume suitable data wherever necessary.

		Marks	B. L	CO's
Q.1	**Attempt any two**	**16**		
A	Define process and explain process states in details with diagram.	**8**	2	1
B	Describe the fields in a process control block (PCB). What is switching overhead?	**8**	2	1
C	List the main difference and similarities between threads and process.	**8**	1	1
Q.2	**Attempt any two**	**16**		
A	With a diagram, explain various communication models for Inter-process communication.	**8**	2	1
B	Consider the following set of processes, with the length of CPU burst time given in milliseconds:	**8**	3	2

Process	Burst Time	Priority	Arrival Time
P1	4	1	0
P2	3	3	1
P3	5	2	2
P4	2	5	3
P5	6	4	4

a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-pre-emptive priority (a larger priority number implies a higher priority), and RR (quantum = 3).
b) Calculate the average waiting and turnaround time for each process.

Which of the algorithms results in the minimum average waiting time (over all processes)?

C What are critical sections? Why mutual exclusion is required? Explain any 2 methods of achieving mutual exclusion in detail. **8** 2 3

Q.3 Attempt any three **18**

A What is dinning philosopher problem? Illustrate its solution with monitor. **6** 2 3

B Infer Bounded Buffer problem with appropriate algorithms. **6** 2 3

C With an example, demonstrate deadlock detection algorithm. **6** 3 3

D Briefly explain the following terms with reference to CPU scheduling **6** 2 4

a) Scheduler
b) Dispatcher

Q.4 Attempt any two **16**

A Define deadlock. What are the necessary conditions of for the deadlock to occur? Explain. **8** 2 3

B Consider the following snapshot of a system: **8** 3 3

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	2	0	0	1	4	2	1	2	3	3	2	1
P1	3	1	2	1	5	2	5	2				
P2	2	1	0	3	2	3	1	6				
P3	1	3	1	2	1	4	2	4				
P4	1	4	3	2	3	6	6	5				

Answer the following questions using Banker's algorithm.

a) What is the content of matrix Need?
b) Is the system is in a safe state?

If the request from the process P1 arrives for (0,1,1,1) can the request be granted immediately?

C Define Paging. Outline the address translation scheme with Paging Hardware diagram. **8** 2 4

Q.5 Attempt any two **16**

A Explain Inverted Page Table with a diagram. **8** 2 4

B With a neat diagram, explain the concept of swapping. Why swapping is necessary? **8** 2 4

C Define virtual memory. Demonstrate with a diagram, how virtual memory is larger than the physical memory. **8** 3 4

Q.6 Attempt any three **18**

A Consider the following page reference string: **6** 3 4

0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 4, 5, 6, 7

How many page faults would occur for the following replacement algorithms, assuming three frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each.

a) FIFO replacement
b) LRU replacement
c) Optimal replacement

B	Define the following terms related to mass storage systems: a) Transfer rate b) Latency c) Seek time d) Head crash	**6**	2	5
C	Consider the following disk request sequence for a disk with 0 to 200 tracks: 176, 79, 34, 60, 92, 11, 41, 114 Head pointer starting at 50 and moving in left direction. Find the number of head movements in cylinders using FCFS, SSTF, C-SCAN scheduling algorithms.	**6**	3	5
D	Write a short note on: a) Storage Area Networks (SAN) b) Network Attached Storage (NAS)	**6**	2	5

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